

Uday Charan
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Professional Summary:

- Senior DevOps and Cloud Engineer with extensive experience architecting, automating, and operating secure, scalable enterprise infrastructure across AWS, Azure, GCP, and on-premises environments.
- Strong expertise in Kubernetes, Amazon EKS, OpenShift, Docker, Helm, Argo CD, and GitOps, with hands-on experience in cluster management, upgrades, node pools, autoscaling, namespaces, and production workload troubleshooting.
- Extensive experience with IaC and configuration automation using Terraform, Terragrunt, CloudFormation, Ansible, and scripting with Python, Bash, PowerShell, and Groovy.
- Designed and maintained enterprise CI/CD pipelines using Jenkins, GitLab CI, AWS CodePipeline, Git, and GitHub, integrating automated build, testing, deployment, and rollback processes.
- Implemented DevSecOps practices by integrating SonarQube, Fortify, Black Duck, Sysdig, Veracode, HashiCorp Vault, RBAC, Kyverno, WAF, network policies, and container security controls into delivery workflows.
- Hands-on experience with AWS services including EC2, S3, IAM, VPC, ELB, Auto Scaling, Route 53, CloudFront, CloudWatch, and EKS, along with experience supporting Azure and GCP cloud environments.
- Implemented advanced deployment strategies including blue-green, canary, zero-downtime deployments, automated rollback, Helm-based releases, and Argo Rollouts for highly available production applications.
- Built and supported enterprise observability solutions using Prometheus, Grafana, ELK/EFK, OpenTelemetry, Datadog, Dynatrace, and CloudWatch for infrastructure, application, and Kubernetes monitoring.
- Experienced in designing secure cloud and Kubernetes environments using IAM, secrets management, zero-trust principles, network security, and compliance-focused controls.
- Leveraged Generative AI and AI-assisted engineering tools including GitHub Copilot, Amazon CodeWhisperer, ChatGPT, and MCP workflows to accelerate infrastructure automation, scripting, troubleshooting, and operational efficiency.
- Strong production support experience handling P1/P2 incidents, change management, hotfix deployments, root-cause troubleshooting, SLA adherence, and ServiceNow/JIRA-based operational processes.
- Proven ability to evaluate emerging cloud, Kubernetes, automation, and AI technologies through POCs and introduce solutions that improve scalability, reliability, security, and engineering productivity.

Technical Skills:

Cloud Platforms:	AWS, Microsoft Azure, Google Cloud Platform (GCP), OpenStack
AWS Services:	EC2, EKS, ECS, S3, RDS, VPC, IAM, Route 53, ELB/ALB, Auto Scaling, CloudFront, CloudWatch, CloudTrail, AWS Config, AWS KMS, Secrets Manager, Lambda, Step Functions.
Azure Services:	AKS, ACR, Azure DevOps, Virtual Networks, Virtual Machines, Storage, Load Balancers
GCP Services:	Cloud Storage, Cloud Functions, Pub/Sub, Memorystore, Cloud Run, OCR Services
Containers & Orchestration	Kubernetes, Amazon EKS, Azure AKS, OpenShift, Docker, Helm, Argo CD, Argo Rollouts, Kubernetes RBAC, Ingress, ConfigMaps, Secrets, StatefulSets, GitOps
Infrastructure as Code & Configuration Management:	Terraform, Terragrunt, AWS CloudFormation, Ansible, OpenStack.
CI/CD & Build Tools:	Jenkins, GitLab CI, GitHub Actions, Azure DevOps, AWS CodePipeline, Maven, JFrog Artifactory
Source Code Management:	Git, GitHub, GitLab, Bitbucket, Subversion
DevSecOps & Security:	SonarQube, Fortify, Black Duck, Sysdig, Veracode, 42Crunch, HashiCorp Vault, Kyverno, IAM, RBAC, WAF, TLS, Network Policies, Secrets Management
Monitoring, Logging & Observability:	Prometheus, Grafana, ELK/EFK, OpenTelemetry, Datadog, Dynatrace, New Relic, CloudWatch, Sumo Logic, Kibana, Logstash, Zabbix
Automation & Scripting:	Python, Bash/Shell, PowerShell, Groovy, YAML, Boto3
Operating Systems:	Linux, Windows
Web & Application Servers:	Apache, Nginx, Tomcat, WebSphere, WebLogic
AI / GenAI:	GitHub Copilot, Amazon CodeWhisperer, ChatGPT, MCP, GenAI Platform Support
DevOps Practices:	CI/CD, Infrastructure as Code (IaC), GitOps, DevSecOps, Blue-Green Deployment, Canary Deployment, Rolling Deployment, Zero-Downtime Deployment, Automated Rollback
ITSM & Collaboration:	ServiceNow, JIRA, Incident Management, Problem Management, Change

	Management, RCA , SLA
Architecture & Platforms:	Microservices, Serverless Architecture, Event-Driven Architecture, Multi-Cloud, Hybrid Cloud, Containerized Platforms

Professional Experience:

JPMC, Remote Sr. DevOps Engineer

May 2025 – Till Date

Responsibilities:

- Designed and implemented secure, highly available cloud infrastructure for enterprise banking and financial applications across AWS and Azure environments using Terraform and Infrastructure as Code [IaC] practices.
- Architected AWS infrastructure using VPC, private/public subnets, NAT Gateways, Application Load Balancers, Auto Scaling Groups, Route 53, IAM, AWS KMS, and Secrets Manager to support secure financial workloads.
- Built and managed Kubernetes platforms using Amazon EKS and AKS, supporting application deployments, scaling, cluster upgrades, pod troubleshooting, resource management, and production availability.
- Developed reusable Terraform modules for networking, compute, storage, security, Kubernetes, and multi-environment provisioning, enabling standardized infrastructure deployment across DEV, SIT, UAT, and PROD banking environments.
- Designed and maintained CI/CD pipelines using Jenkins, Azure DevOps, GitHub Actions, Git, and YAML to automate application builds, testing, security validation, container image creation, and production deployments.
- Implemented GitOps-based Kubernetes deployments using Argo CD and Helm, improving deployment consistency, traceability, configuration control, and rollback capabilities for financial applications.
- Implemented cloud and Kubernetes security controls using IAM least-privilege access, RBAC, encryption, Secrets Manager, HashiCorp Vault, security groups, network policies, and centralized logging.
- Integrated security and code-quality validation into CI/CD pipelines using SonarQube, Sysdig, Fortify, Black Duck, and related scanning tools to support DevSecOps and enterprise compliance requirements.
- Implemented blue-green, canary, rolling, and zero/minimal-downtime deployment strategies to reduce application disruption during critical banking production releases.
- Built monitoring and observability capabilities using Prometheus, Grafana, CloudWatch, Datadog/Dynatrace, and centralized logging to monitor application availability, infrastructure health, performance, and capacity.
- Automated infrastructure provisioning, configuration management, patching, and operational activities using Terraform, Ansible, Python, and Bash/Shell, reducing manual effort and improving environment consistency.
- Collaborated with application, security, networking, database, QA, and infrastructure teams during architecture reviews, production readiness, release planning, troubleshooting, and change implementation for financial platforms.
- Supported P1/P2 production incidents, performed RCA, and participated in hotfix deployments, rollback activities, post-deployment validation, and ServiceNow-based change management.
- Created technical documentation, deployment procedures, operational runbooks, disaster recovery procedures, and knowledge-transfer documentation to support reliable financial production operations.

Environment: AWS, Azure, EKS, AKS, VPC, EC2, S3, IAM, AWS KMS, Secrets Manager, Route 53, ALB, Auto Scaling, CloudWatch, Azure DevOps, ACR, Terraform, Ansible, Jenkins, GitHub Actions, Git, Argo CD, Helm, Docker, Kubernetes, Python, Bash/Shell, Linux, Prometheus, Grafana, SonarQube, Sysdig, Fortify, Black Duck, ServiceNow, JIRA, CI/CD, IaC, GitOps, DevSecOps.

CVS, Remote Sr. DevOps Engineer

Feb 2024 - Apr 2025

Responsibilities:

- Designed, implemented, and supported secure, scalable cloud platforms for healthcare and clinical-trial applications across AWS and Azure environments, focusing on availability, automation, and consistent infrastructure delivery.
- Managed AWS EKS and Kubernetes environments supporting containerized healthcare applications, including deployments, services, configurations, secrets, scaling, resource utilization, and production troubleshooting.
- Developed reusable Terraform modules to automate provisioning of cloud compute, networking, storage, identity, and Kubernetes infrastructure across development, testing, pre-production, and production environments.
- Designed and enhanced CI/CD pipelines using GitHub Actions, Jenkins, GitLab CI, and Git to automate application builds, testing, container image creation, security validation, environment promotion, and deployments.
- Implemented Argo CD and GitOps-based deployment practices for Kubernetes workloads, providing controlled, version-based, traceable, and repeatable application releases across healthcare environments.
- Built and maintained Docker images and Helm-based Kubernetes deployments to standardize application packaging

and improve consistency across multiple environments.

- Engineered automated AWS account and environment provisioning workflows using Terraform, AWS Step Functions, Lambda, and Python/Boto3, reducing manual infrastructure setup and improving platform scalability.
- Supported multi-account AWS environments with centralized governance, cross-account IAM roles, access controls, security baselines, and standardized infrastructure provisioning.
- Implemented cloud and Kubernetes security controls including IAM, RBAC, secrets management, TLS, namespace isolation, container security policies, and secure configuration baselines.
- Established platform monitoring and observability using Prometheus, Grafana, Amazon CloudWatch, New Relic, application logs, and centralized logging to proactively identify availability, performance, and infrastructure issues.
- Led troubleshooting of complex EKS/Kubernetes issues involving pod failures, CPU/memory constraints, persistent storage, application connectivity, configuration, and deployment failures.
- Implemented rolling, canary, and zero/minimal-downtime deployment approaches with rollback capabilities to improve release reliability and reduce disruption to healthcare application environments.
- Automated operational and infrastructure activities using Terraform, Ansible, Python, and Bash/Shell, reducing configuration drift, repetitive administration, and manual deployment activities.
- Collaborated with development, QA, database, security, network, and application-support teams on application upgrades, infrastructure changes, production releases, and platform improvements.
- Identified cloud optimization opportunities through resource utilization, storage, snapshot, capacity, and configuration reviews, improving infrastructure efficiency and controlling AWS operational costs.

Environment: AWS, Azure, Amazon EKS, EC2, IAM, AWS Lambda, Step Functions, CloudWatch, Kubernetes, Docker, Helm, Argo CD, GitOps, Terraform, Ansible, GitHub Actions, Jenkins, GitLab CI, Git, Python, Boto3, Bash/Shell, Linux, Prometheus, Grafana, New Relic, RBAC, TLS, CI/CD, IaC, DevSecOps, Microservices, Centralized Logging.

Optum Health, CA
DevOps Engineer

Aug 2022 – Jan 2024

Responsibilities:

- Designed, implemented, and supported cloud-native DevOps platforms across AWS, GCP, and Azure, enabling scalable application, data-processing, and AI-driven workloads across multiple environments.
- Built and managed event-driven healthcare data-processing pipelines on GCP, automating document ingestion, preprocessing, OCR, entity extraction, and comparison workflows against EMR datasets.
- Designed event-driven workflows using Google Cloud Storage events, Cloud Functions, Pub/Sub, and Memorystore to provide reliable asynchronous processing, caching, and state management across distributed pipeline components.
- Containerized application and data-processing services using Docker and deployed workloads through GCP Cloud Run, enabling scalable serverless execution and consistent deployments across environments.
- Designed, deployed, and supported AWS containerized platforms using Amazon EKS, ECS, Docker, Helm, EC2, S3, and RDS to support scalable application and GenAI workloads.
- Managed Kubernetes environments including namespaces, application deployments, container scaling, configuration management, resource utilization, and production troubleshooting.
- Built and maintained CI/CD pipelines using Jenkins, integrating application builds, testing, security scanning, code-quality validation, container image creation, artifact management, and environment deployments.
- Managed Docker image repositories using JFrog Artifactory and integrated Sysdig container-image scanning into delivery pipelines to identify security vulnerabilities before deployment.
- Integrated SonarQube and 42Crunch static/dynamic API security scanning into CI/CD workflows, strengthening code-quality and DevSecOps controls across application releases.
- Provisioned cloud infrastructure using Terraform and reusable IaC practices, enabling consistent and repeatable infrastructure deployment across cloud environments.
- Conducted cost, operational, performance, and quality assessments for GCP-to-Azure migration, developing benchmark reports to support cloud-platform migration decisions.
- Provisioned equivalent Azure infrastructure using Terraform to replicate event-driven storage, compute, processing, and database components while maintaining platform reliability and cost efficiency.
- Established monitoring, logging, and observability using Amazon CloudWatch, CloudTrail, AWS Config, ELK, and application-level monitoring to proactively identify availability and operational issues.
- Automated operational recovery activities, including service and node restart workflows, to reduce manual intervention and improve application availability.
- Mentored engineers on Terraform, IaC, Azure adoption, cloud migration, benchmark testing, and standardized DevOps practices while collaborating with development, QA, architecture, and operations teams.

Environment: AWS, GCP, Azure, Amazon EKS, ECS, EC2, S3, RDS, CloudWatch, CloudTrail, AWS Config, Google Cloud Storage, Cloud Functions, Pub/Sub, Memorystore, Cloud Run, GCP OCR Services, Terraform, Kubernetes, Docker, Helm, Jenkins, JFrog Artifactory, Sysdig, SonarQube, 42Crunch, ELK, Python, Git, CI/CD, IaC, DevSecOps, Serverless, Event-Driven Architecture, EMR Data.

Lincoln Financial Group, Philadelphia, PA
DevOps Engineer / Cloud DevOps Engineer

Jul 2020 – Dec 2022

Responsibilities:

- Maintained the overall SVN architecture that includes setting up branching process, setting up user accounts, and maintaining the user access across the organization as an admin.
- Worked on Multi project jobs in Jenkins and used pipeline plugin to manage them.
- Configured the Backup Management in Jenkins for Config. xmls for all the jobs using plugin named periodic backup.
- Performed integration of Code Quality Analysis Techniques like Check styles, Find Bugs, and PMD with CI tools like Hudson/Jenkins.
- Experienced in authoring pom.xml files, performing releases with the Maven release plugin, Mavenization of Java projects and managing Maven repositories.
- Worked with development team to migrate Ant scripts to Maven.
- Involved in Setting up Continuous Integration Environment using Jenkins and responsible for design and maintenance of the GIT Repositories, views, and the access control strategies.
- Worked on Maven to create artifacts from source code and deploy them in Nexus central repository for internal deployments.
- Set up continuous integration and formal builds using Jenkins with Maven and Tomcat repository.
- Configured servers to provide networking services, including HTTP/HTTPS, File Transfer Protocol (FTP), network file system (NFS), and server message block (SMB), Simple Mail Transfer Protocol (SMTP), secure shell (SSH) and Network Time Protocol (NTP).
- Installed and configured DHCP, web (Apache, IIS), mail (SMTP, IMAP, POP3), and file servers on Linux and Windows servers.
- Worked on designing and deploying a multitude application utilizing almost all the main services of the AWS stack (like EC2, S3, RDS, VPC, IAM, ELB, Cloud watch, Route 53, Lambda and Cloud Formation) focused on high availability, fault tolerance environment.
- Created Cloud Formation Template for main services like EC2, VPC and S3 for reuse the current environment. Created network architecture on AWS VPC, subnets, Internet Gateway, Route Table and NAT Setup.
- Developed a Python Scripts to manage few services on AWS using SDK BOTO.
- Designed high availability environment for Application servers and database servers on EC2 by using ELB and Auto-scaling.
- Integrated Jenkins with GitHub private repositories builds Automation tools (Maven and Ant), and Artifact repository for pushing successful build code.
- Configured NAGIOS monitoring tool to maintain the number of servers on different environments.

Environment: Jenkins, Maven, Hudson, Ant, Puppet, Tomcat, HTTP, FTP, NFS, SMB, SMTP, SSH, NTP, DHCP, EC2, VPC, S3, IAM, ELB, Auto Scaling, CloudWatch, JSON, Python script, Shell script, GIT, GitHub, Puppet, Nagios and Linux servers

Tech Mahindra, India
Infrastructure & DevOps Engineer — Private Cloud

Nov 2018 – Jun 2020

Responsibilities:

- Supported enterprise cloud and container platforms across development, test and production environments.
- Designed and maintained CI/CD pipelines using Jenkins, Git and Azure DevOps.
- Automated application builds, package creation, containerisation and deployment.
- Created Dockerfiles and optimised container images for production workloads.
- Managed Kubernetes and OpenShift application deployments.
- Created Helm charts and Kubernetes manifests for repeatable application deployments.
- Managed Kubernetes namespaces, RBAC, secrets, resource limits, services and ingress.
- Supported Kubernetes cluster upgrades, node maintenance and application troubleshooting.
- Implemented health checks, readiness probes and liveness probes for containerised applications.
- Troubleshooting Kubernetes scheduling, networking, storage and container runtime issues.
- Automated infrastructure and configuration management using Ansible.
- Developed Bash and Python scripts for operational automation.

- Managed Linux servers and performed system-level troubleshooting.
- Worked with VMware virtualisation environments and enterprise infrastructure.
- Supported OpenStack-based private cloud environments including compute, networking and storage services.
- Implemented monitoring using Prometheus, Grafana, Zabbix and ELK.
- Troubleshooting CPU, memory, disk, network, application and Kubernetes performance issues.
- Supported production incidents and performed detailed RCA.
- Collaborated with application teams to onboard workloads into Kubernetes.
- Implemented standardised deployment templates and operational procedures.
- Participated in disaster recovery, capacity planning and platform reliability activities.

Innominds Software, India
Build Engineer

July 2017 – Oct 2018

Responsibilities:

- Setting up the new build environment for various products in Linux environment Implemented the Release practice and responsible for pushing builds into DEV / UAT/Pre-Production/ Production
- Build Pipeline design and optimization: GIT, Subversion, Maven, Jenkins and artifactory for J2EE application deployments.
- Involved in Deployment of the application using Tomcat.
- Worked on Jenkins to configure distributed builds. Worked on installing plugging, creating jobs, integrating test cases.
- Managing/Tracking the defects status using jira
- Daily activities included delivering daily/nightly builds for the scrum teams, supporting teams with SCM requests, maintaining code stability using CI methodologies, code merging, deployments and troubleshooting other day-to-day issues.
- Inputs to Planning and estimation for configuration and release management activities. This involves exploring new tools and plan for release & configuration management process designing and implementations
- Co-work with the globally distributed Development team to fix the Build related issues
- Promoting code from one Environment to another Environment
- Perform application configurations review, code review and impact assessment before the code deployment in production servers to avoid any impact on business.
- Perform detailed application and web layer log analysis using tools like Sumologic, Kibana/logStash, Grafana to analyze technical issues
- Configured Ansible modules for infrastructure automation. Worked with Ansible playbooks for virtual and physical instance provisioning.
- Handling configuration management, patching, and software deployments through Ansible
- Worked in all areas of Jenkins setting up CI for new branches, Build Automation, Plugin Management and Securing Jenkins and setting up master/slave configurations.
- Used Kubernetes to provide a platform for automating deployment, scaling and operations of application containers across clusters.
- Building and deploying under different testing environments starting from development to Pre-Prod.
- Make sure the dev/sit/uat environments are always up and running.
- Monitoring the application servers and troubleshooting/fixing if any issues.
- Maintaining gitlab, Jenkins, sonar and Fortify tools.
- Worked on CrashLoopBackoff debug, troubleshooting Kubernetes Pod logs and debugging pod restart and resolving all pod issues in production.
- Experience in installing and configuring the application and web servers like Apache, Nginx, Tomcat, WebSphere, WebLogic and deploying several applications on the servers
- Utilize analytical and critical thinking to identify, analyze and understand technical issues providing solutions to Client / Advisor
- Tracking and providing timely resolution for incidents / Problem tickets within the defined SLA.

Environment: Tomcat, WebSphere, Blackduck, Sumologic, Sonar, Git, AWS, Windows, Linux, Kubernetes, Ansible, Fortify, Maven, CA, Jira, Marathon, Udeploy, ServiceNow.